

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2021 P2H Worked Solutions

May 2021 | P2H

27 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May 2021 P2H - May 2021 | P2H

- 1** Write 600 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Fractions**

May 2021 P2H - May 2021 | P2H

2 Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Fractions**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May 2021 P2H - May 2021 | P2H

- 3** The bearing of Paris from London is 149°

Work out the bearing of London from Paris.

o

(Total for Question 3 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 3

Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P2H - May 2021 | P2H

4 $\mathcal{E} = \{\text{letters of the alphabet}\}$ $B = \{\text{b, r, a, z, i, l}\}$ $I = \{\text{i, r, e, l, a, n, d}\}$

(a) List the members of the set

(i) $B \cup I$ (ii) $B \cap I'$
(2) $K = \{\text{k, e, n, y, a}\}$ Cody writes down the statement $B \cap K = \emptyset$

Cody's statement is wrong.

(b) Explain why.

.....
(1).....
(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

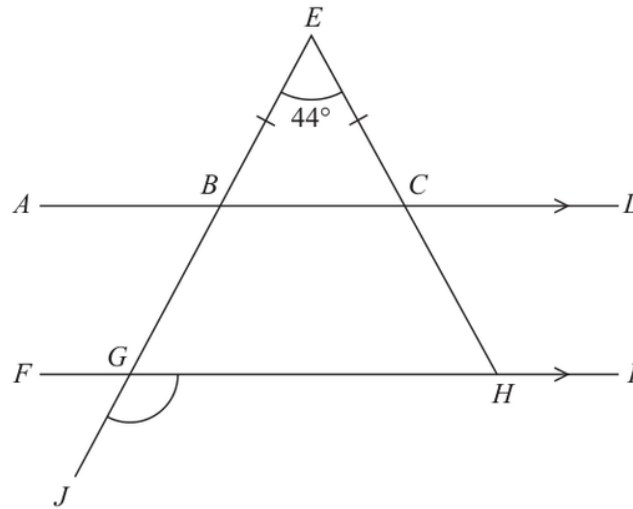
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

May 2021 P2H - May 2021 | P2H

5Diagram **NOT**
accurately drawn

$ABCD$ and $FGHI$ are parallel straight lines.
 $EBGJ$ and ECH are straight lines.

$BE = CE$
 Angle $BEC = 44^\circ$

Work out the size of angle JGH .
 Give a reason for each stage of your working.

o

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 5

TOPIC: **Angles in Polygons & Parallel Lines**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Percentages**

May 2021 P2H - May 2021 | P2H

6 Mariana sells bags of bird food.

The bags that Mariana sold last week each contained 12 kg of seeds.

The bags that she is going to sell next week will each contain a mixture of nuts and seeds where for each bag

$$\text{weight of nuts : weight of seeds} = 4 : 5$$

The total weight of the nuts and the seeds in each bag will be 19.35 kg

The weight of seeds in each bag that Mariana sells next week will be less than the weight of seeds in each bag that Mariana sold last week.

Work out this decrease as a percentage of the weight of seeds in each bag that Mariana sold last week.

Give your answer correct to one decimal place.

..... %

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: **Percentages**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

May 2021 P2H - May 2021 | P2H

7 Here is a right-angled triangle.

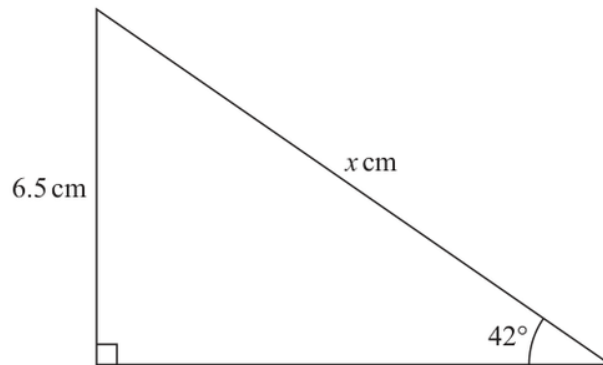


Diagram **NOT**
accurately drawn

Work out the value of x .
Give your answer correct to one decimal place.

$x =$

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

8 Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

$$a = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: **Simultaneous Equations**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Factorising**

May 2021 P2H - May 2021 | P2H

9 (i) Factorise $x^2 + 2x - 24$
(2)(ii) Hence solve $x^2 + 2x - 24 = 0$
(1)

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Factorising**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

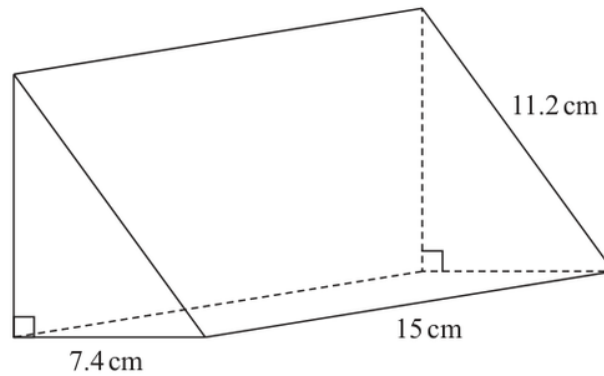
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Volume & Surface Area**

May 2021 P2H - May 2021 | P2H

10 Here is a triangular prism.Diagram **NOT**
accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

..... cm³

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: **Volume & Surface Area**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Compound Interest & Depreciation**

May 2021 P2H - May 2021 | P2H

11 Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

- (a) Work out how much Chengbo paid for the house.
Give your answer correct to 3 significant figures.

..... yuan
(3)

Zhi bought a house on 1st January 2017
When she bought the house, its value was 120 000 yuan.

The value of the house increased by 1.8% per year.

- (b) Work out the value of Zhi's house on 1st January 2020
Give your answer correct to 3 significant figures.

..... yuan
(3)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 6

TOPIC: **Compound Interest & Depreciation**

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No worked solution is saved yet.

EA

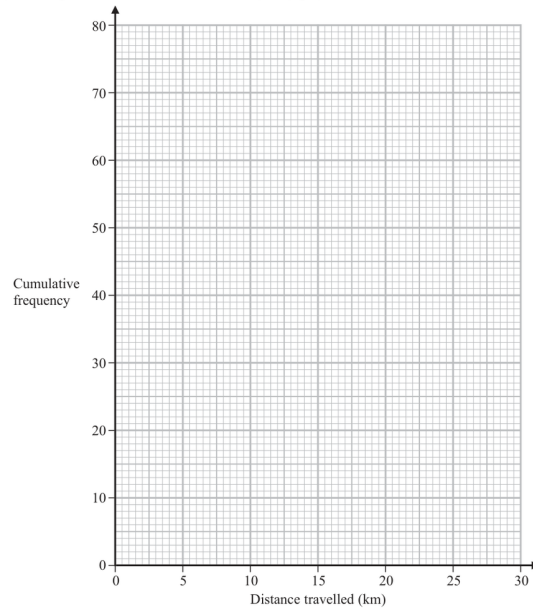
PAST PAPER QUESTION**Q#: 12****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

May 2021 P2H - May 2021 | P2H

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km

(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km

(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

1

2

(2)

(Total for Question 12 is 7 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

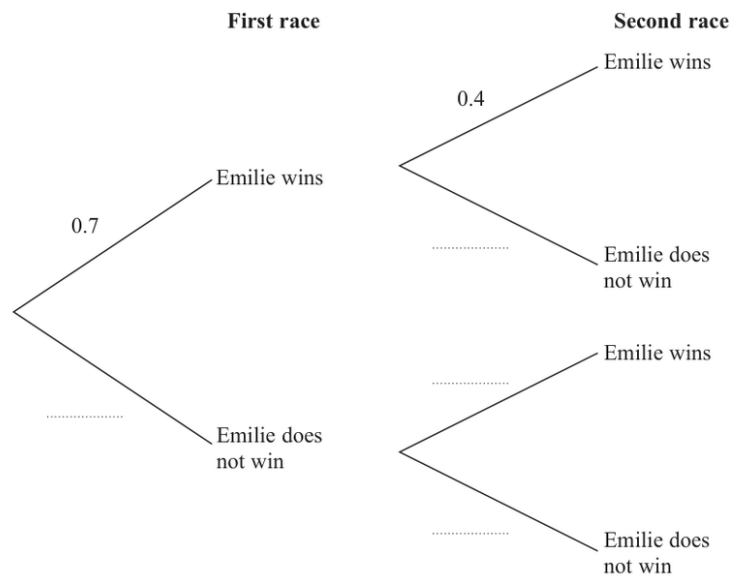
PAST PAPER QUESTION**Q#: 13****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May 2021 P2H - May 2021 | P2H

13 Emilie takes part in two races.

The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

(3)

(Total for Question 13 is 8 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 8

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Algebraic Roots & Indices**

May 2021 P2H - May 2021 | P2H

14 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Algebraic Roots & Indices**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Graphs of Functions**

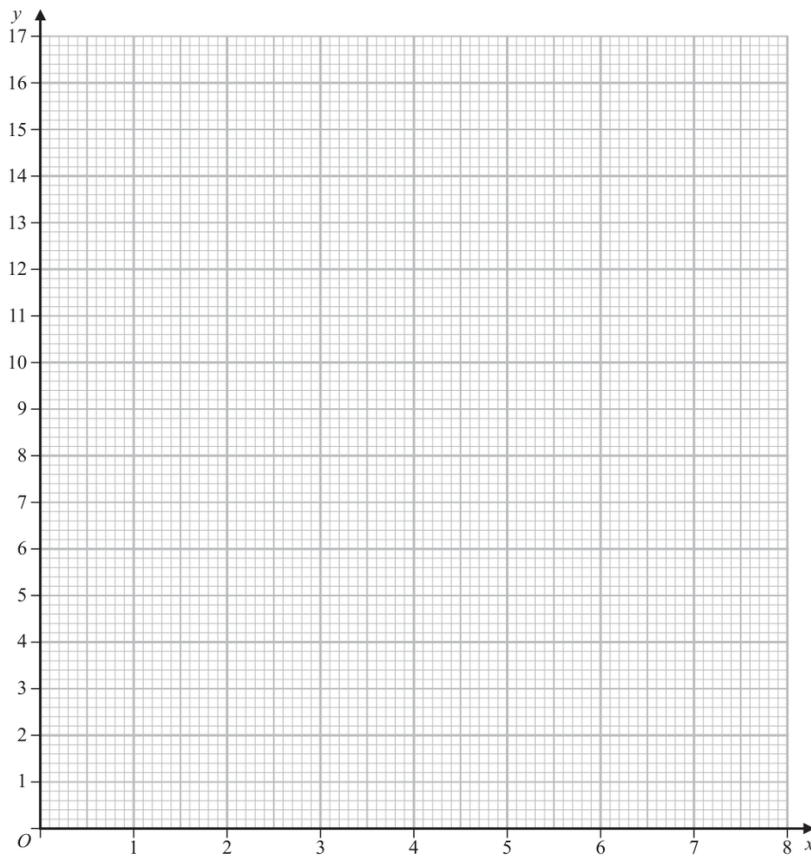
May 2021 P2H - May 2021 | P2H

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$



(2)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 4

TOPIC: **Graphs of Functions**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 6****TOPIC: Direct & Inverse Proportion**

May 2021 P2H - May 2021 | P2H

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$$A = \text{.....}$$

(3)

(Total for Question 16 is 6 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 6

TOPIC: **Direct & Inverse Proportion**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Coordinate Geometry**

May 2021 P2H - May 2021 | P2H

17 The straight line **L** passes through the points $(4, -1)$ and $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the y -axis at the point $(0, 8)$

Find the coordinates of the point where **M** intersects the x -axis.

(.....,))

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 4

TOPIC: **Coordinate Geometry**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

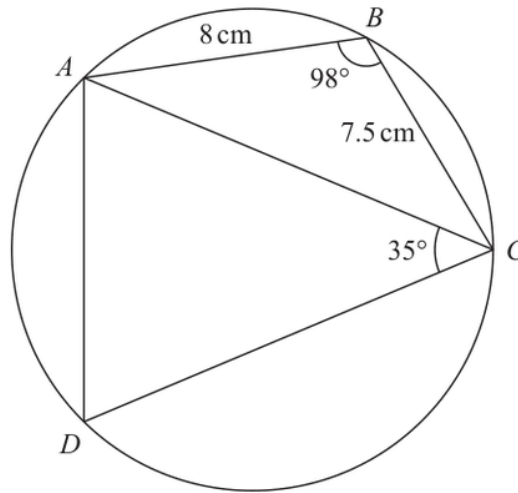
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 6****TOPIC: Circle Theorems**

May 2021 P2H - May 2021 | P2H

18Diagram **NOT**
accurately drawn $ABCD$ is a quadrilateral where A , B , C and D are points on a circle. $AB = 8 \text{ cm}$ $BC = 7.5 \text{ cm}$ Angle $ABC = 98^\circ$ Angle $ACD = 35^\circ$ Work out the perimeter of quadrilateral $ABCD$.

Give your answer correct to one decimal place.

..... cm

(Total for Question 18 is 6 marks)

PAST PAPER SOLUTION

Q#: 18

Marks: 6

TOPIC: **Circle Theorems**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

19 Solve the simultaneous equations

$$\begin{aligned}y &= 3 - 2x \\ x^2 + y^2 &= 18\end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 5

TOPIC: **Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

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Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Differentiation**

May 2021 P2H - May 2021 | P2H

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Differentiation**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

EA

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May 2021 P2H - May 2021 | P2H

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Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Differentiation**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Completing the Square**

May 2021 P2H - May 2021 | P2H

- 22** The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants,
describe fully a series of transformations that map the curve **S** onto the curve **T**.

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Completing the Square**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Completing the Square**

May 2021 P2H - May 2021 | P2H

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describe fully a series of transformations that map the curve **S** onto the curve **T**.

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Completing the Square**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P2H - May 2021 | P2H

23 Pippa has a box containing N pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N .

Show clear algebraic working.

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Probability Toolkit**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P2H - May 2021 | P2H

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The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N .

Show clear algebraic working.

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Probability Toolkit**

May 2021 P2H - May 2021 | P2H

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No worked solution is saved yet.

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